Microscopic Image Enhancement and Segmentation of Dry Foods

Rubina Erin
220218025

Zeynep Ruveyda Demirci
220218012

Classical image processing for clearer microscopic inspection before packaging

PROJECT PURPOSE & APPLICATION

Dry food samples may contain small surface structures that are difficult to distinguish in raw microscope images. This project improves their visibility and separates contrasting regions for visual inspection during quality control, without identifying or classifying any material.

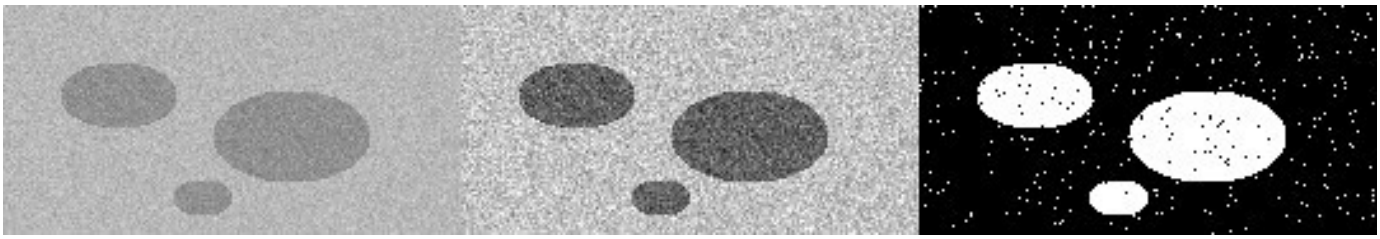
IMAGE ACQUISITION

Capture magnified images of dry food samples using an optical microscope under controlled lighting and consistent magnification. Comparable imaging conditions support reliable before-and-after visual comparison.

CLASSICAL IMAGE PROCESSING WORKFLOW

01 RGB-to-HSI Conversion	Separate Hue (H), Saturation (S), and Intensity (I) to examine color and brightness independently.
02 Histogram Equalization	Redistribute intensity values to improve contrast and make subtle surface details easier to see. $s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$
03 Otsu Thresholding	Choose a histogram-based threshold for the enhanced intensity channel and generate a binary mask. $T^* = \arg\max_t [\omega_0(T)\,\omega_1(T)\,(\mu_0(T)-\mu_1(T))^2]$
04 Morphological Opening	Reduce small isolated foreground pixels using erosion followed by dilation. $A \circ B = (A \ominus B) \oplus B$
05 Morphological Closing	Fill narrow gaps and connect nearby mask regions using dilation followed by erosion. $A \bullet B = (A \oplus B) \ominus B$
06 Visual Comparison	Compare the original, enhanced, and segmented images to observe how clearly surface structures are represented.

EXPECTED OUTPUT / ILLUSTRATIVE PROCESSING EXAMPLE



Original Image **Enhanced Image** **Binary Mask**

LIMITATION

A binary mask represents intensity-based regions, not confirmed mold. Different food textures or particles may appear similar. The examples are illustrative, not experimental results.